



MASTG

- MASTG-TEST-0035: Testing for Overlay Attacks
- MASTG-TEST-0037: Testing WebViews Cleanup
- MASTG-TEST-0250: References to Content Provider Access in WebViews
- MASTG-TEST-0251: Runtime Use of Content Provider Access APIs in WebViews
- MASTG-TEST-0252: References to Local File Access in WebViews
- MASTG-TEST-0253: Runtime Use of Local File Access APIs in WebViews

L1

L2

ANDROID

MASWE-0069

TEST



MASTG-TEST-0251: Runtime Use of Content Provider Access APIs in WebViews

Overview

This test is the dynamic counterpart to [References to Content Provider Access in WebViews](#).

Steps

- Run a dynamic analysis tool like [Frida \(Android\)](#) and either:
 - enumerate instances of `WebView` in the app and list their configuration values
 - or explicitly hook the setters of the `WebView` settings

Observation

The output should contain a list of `WebView` instances and corresponding settings.

Evaluation

Fail:

The test fails if all of the following are true:

- `JavaScriptEnabled` is `true`.
- `AllowContentAccess` is `true`.
- `AllowUniversalAccessFromFileURLs` is `true`.

You should use the list of content providers obtained in [References to Content Provider Access in WebViews](#) to verify if they handle sensitive data.

Note: `AllowContentAccess` being `true` does not represent a security vulnerability by itself, but it can be used in combination with other vulnerabilities to escalate the impact of an attack. Therefore, it is recommended to explicitly set it to `false` if the app does not need to access content providers.

Pass:

The test passes if any of the following are true:

- `JavaScriptEnabled` is `false`.
- `AllowContentAccess` is `false`.
- `AllowUniversalAccessFromFileURLs` is `false`.

Best Practices

- MASTG-BEST-0011: Securely Load File Content in a WebView
- MASTG-BEST-0012: Disable JavaScript in WebViews
- MASTG-BEST-0013: Disable Content Provider Access in WebViews

Demos

- MASTG-DEMO-0030: Uses of WebViews Allowing Content Access with Frida